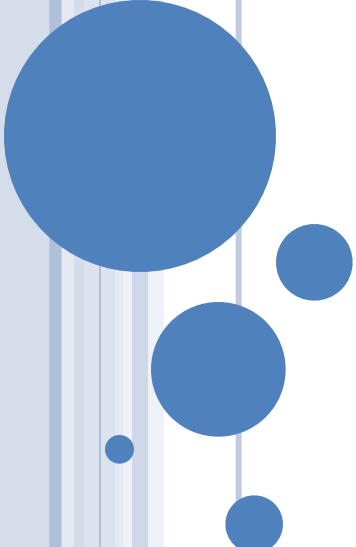


https://hackmd.io/6_KUxQtRQr6H48S7U5OL1Q?view

INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 9 Large-Sample Tests of Hypotheses

https://colab.research.google.com/drive/1TrMHmr5xo5a9-9y6O5t_NM6_zpu9ODyv#scrollTo=kA_ZQISj2nw-

BRAIN-STORMING

- Suppose you draw a fair coin for 100 times.
- Do you think it is common to observe 10 heads?
- Do you think it is common to observe 55 heads?
- Why or why not?



KEY INGREDIENTS

- Model specification: $\mathcal{F}(\theta)$, and want to know θ from your scenario.
- Find an **estimator** of θ
- Standardize the estimator to obtain a **statistic**, and identify its sampling distribution
- Then, we can do two statistical approaches
 - Confidence interval (Ch 8)
 - Hypothesis test (Ch 9 and after)



A decorative graphic on the left side of the slide. It features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The circles are arranged in a way that they appear to be floating or overlapping the stripes.

9.1 HYPOTHESIS TESTS

H_0 VS H_a

- H_0 , null hypothesis: a contradiction of the alternative hypothesis
 - $H_0 : \theta = \theta_0$
- H_a (or H_1), Alternative hypothesis: the hypothesis the researcher wants to support
 - $H_1 : \theta \neq \theta_0$



ACTIONS

Only two actions:

1. **Reject $H_0 \neq$ accept H_a ,**
Reject $H_0 \approx$ accept H_a
2. **Do not reject $H_0 \neq$ accept H_0**
Do not reject $H_0 \approx$ accept H_0



TYPE I AND TYPE II ERRORS

- **Type I error**
refers to
reject H_0 when
it is in fact
true.
- **Type II error**
refers to
accept H_0
when it is in
fact false.

	H_0 true	H_0 False (H_a true)
Reject H_0	Type I error	Correct action
Do not reject H_0	Correct action	Type II error



- The significance level α is the probability of rejecting H_0 when it is in fact true. Or, the probability of Type I error:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P(\text{Type I error})$$

- β is the probability of accepting H_0 when it is in fact false:

$$\beta = P(\text{accept } H_0 \mid H_0 \text{ false}) = P(\text{Type II error})$$

- The power of the test is $(1 - \beta)$, the probability of rejecting H_0 when it is false.



Imagine you are in an era where computer was not accessible...

1. Check Tables

2. Using python (not in the exam)

人工時期與機械時期計算工具之演進表

	工具名稱	年代(西元)	發明者	說明
人工時期	算盤	約1100年	中國人	人類最早的計算工具
	計算尺	1620	艾德蒙·甘特 (Edmund Gunter)	西方早期常用的計算工具
機械時期	加法器	1642	巴斯卡 (Blaise Pascal)	以齒輪帶動計數轉盤，大小如香煙盒
	自動進位加法器	1673	萊布尼茲 (Gottfried Leibniz)	可連續加、乘
	四則運算機	1820	湯瑪斯 (C.X. Thomas)	可做加、減、乘、除運算
	差分機	1833	巴貝基 (Charles Babbage)	具有現代電腦輸入、輸出處理功能，由於巴貝基對於計算工具貢獻良多，因此被尊稱為計算機之父
	打孔卡片	1890	何樂禮 (Herman Hollerith)	以打孔位置來儲存資料 🥰
	馬克一號計算機 (MARK-1)	1944	艾肯 (Howard Aiken)	史上最大的電力機械計算機 以電磁繼電器做成

The RR guarantees that
 $P(\text{Type I error}) = \alpha$

SOP

1. Set up H_0 vs. H_a .
2. Decide the significant level α (usually $\alpha=0.05, 0.1, 0.01$).
3. Find a test statistic (検定統計量, standardized estimator under H_0) and its sampling distribution.
4. Calculate the realized test statistic from the data.
5. Find (a) the rejection region or (b) the p -value.
6. Conclude



(A) THE REJECTION REGION APPROACH (OR CRITICAL VALUE APPROACH)

5. Find the rejection region

6. Conclusion:

- If the realized statistics falls into the rejection region, reject H_0 .
- Otherwise, do not reject H_0 .



(B) THE P-VALUE APPROACH

5. Calculate the p -value: The p -value is the probability of observing a test statistic as extreme as or more than the one observed assuming H_0 is true.

p -value

= $P(\text{Observing a realized test statistic or more extreme values} | H_0 \text{ true})$

6. Conclusion:

- If p -value is smaller than α , reject H_0 .
- Otherwise, do not reject H_0



SLIDING SCALE

If you are reading a research report, how small should the p -value be before you decide to reject H_0 ? Many researchers use a “sliding scale” to classify their results.

- If the p -value is less than .01, H_0 is rejected. The results are **highly significant**.
- If the p -value is between .01 and .05, H_0 is rejected. The results are **statistically significant**.
- If the p -value is between .05 and .10, H_0 is usually not rejected. The results are only **tending toward statistical significance**.
- If the p -value is greater than .10, H_0 is not rejected. The results are **not statistically significant**.



AN EXAMPLE

Table 5.2: **Static features of selected stocks and exchange rate.** This table provides the static features with a study period from 2008/1/2 to 2021/12/31, including the first four moments, minimum, maximum and the Jarque-Bera statistic.

Stock	mean	sd	skew	kurt	min	max	JB
6505	0.00014	0.0167	0.04	3.26***	-0.1056	0.0945	1518***
2201	0.00003	0.0215	0.34***	3.64***	-0.1047	0.1199	1965***
1303	0.00019	0.0161	-0.14***	3.26***	-0.1024	0.0745	1533***
1301	0.00025	0.0154	-0.04	3.41***	-0.0971	0.0688	1663***

Tables 5.2 and 5.3 provide static and dynamic measures for the selected stocks and the exchange rate, respectively. Here, *, ** and *** denote the significant **level** 10%, 5% and 1%, respectively. Table 5.2 shows the static features, including the first four moments, minimum return, maximum return and the Jarque-Bera statistic. Among them, the mean, skewness and kurtosis are tested for the **significance**. The Jarque-Bera tests whether the given data follows a normal distribution. The results show that all stock returns do not follow a normal distribution under 1% significant **level**.



The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, with smaller circles arranged in a descending pattern below it.

9.2 ONE POPULATION MEAN

IN-CLASS EXERCISE

- $H_0 : p = p_0$ versus $H_a : p > p_0$

Show: if Z^* falls in the RR $\iff$ p-value $< \alpha$

($\Rightarrow$) If Z^* falls in RR $= \{z : z > z_\alpha\}$, then $Z^* > z_\alpha$

$$\text{P-value} = P(Z > Z^*) < P(Z > z_\alpha) = \alpha$$

($\Leftarrow$) If p-value $< \alpha$, $P(Z > Z^*) < \alpha$, $Z^* > z_\alpha$,
 Z^* falls in the RR.



The RR guarantees that $P(\text{Type I error}) = \alpha$

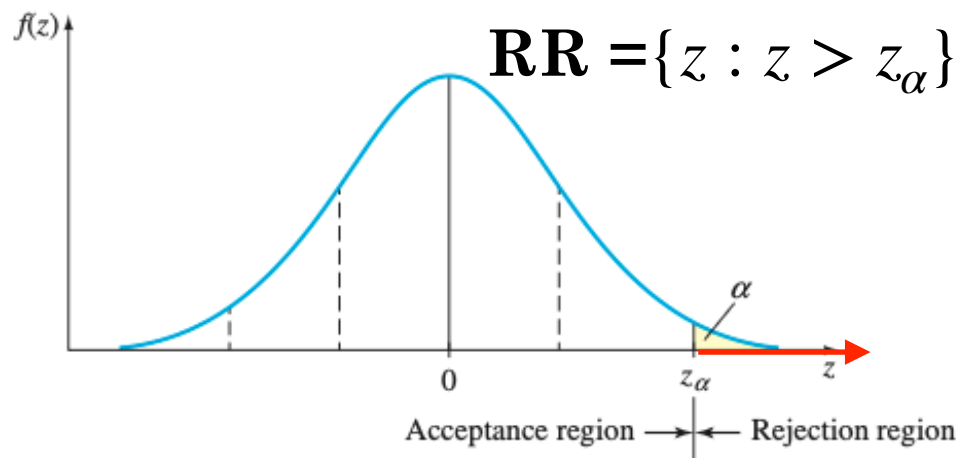
RIGHT-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu \leq \mu_0$ versus $H_a : \mu > \mu_0$
- $H_0 : \mu < \mu_0$ versus $H_a : \mu \geq \mu_0$



Figure 9.3

The rejection region for a right-tailed test with significance level α



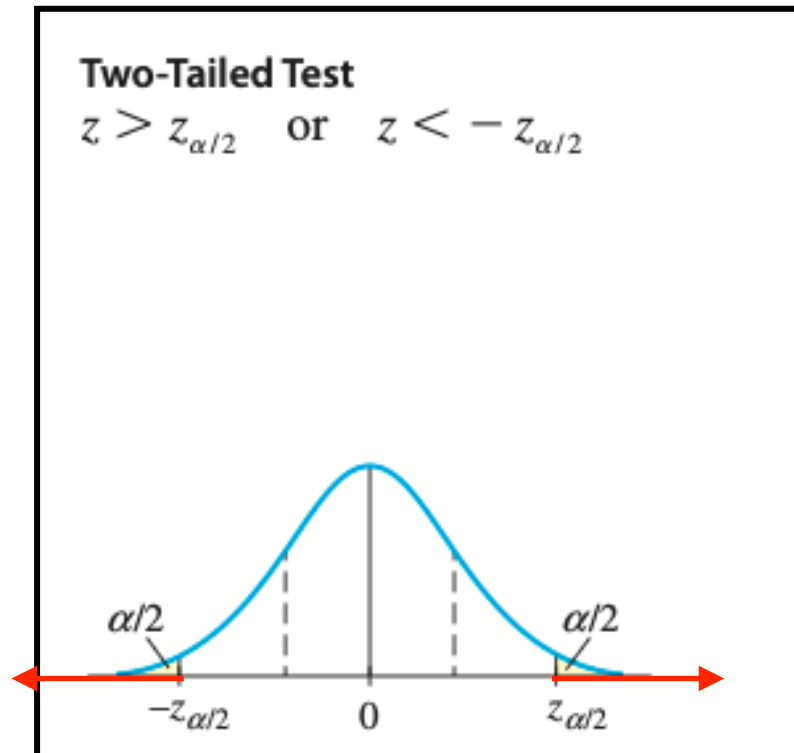
Show: if Z^* falls in the RR $\iff$ p-value $< \alpha$

The RR guarantees that $P(\text{Type I error}) = \alpha$

TWO-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$

$$\mathbf{RR} = \{z : z > z_{\alpha/2}, \text{ or } z < -z_{\alpha/2}\} = \{z : |z| > z_{\alpha/2}\}$$

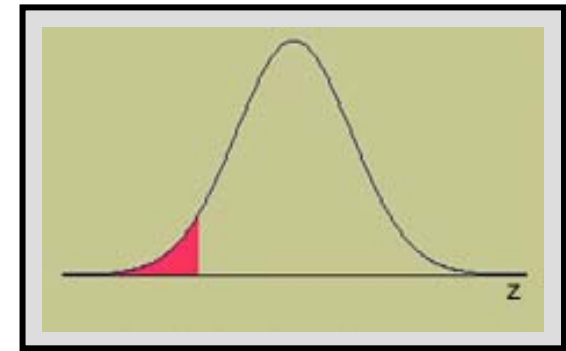


Show: if Z^* falls in the RR $\iff$ p-value $< \alpha$

The RR guarantees that
 $P(\text{Type I error}) = \alpha$

LEFT-SIDED TEST

- $H_0 : \mu = \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu \geq \mu_0$ versus $H_a : \mu < \mu_0$
- $H_0 : \mu > \mu_0$ versus $H_a : \mu \leq \mu_0$



$$\mathbf{RR} = \{z : z < -z_\alpha\}$$

Show: if Z^* falls in the RR $\iff$ p-value $< \alpha$

KEY INGREDIENTS (A1)

- Model: $X_i \sim^{i.i.d.} \mathcal{F}(\mu, \sigma^2)$ for $i = 1, \dots, n$.
- Interest: μ .
- Use $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ to estimate μ .



KEY INGREDIENTS (A2)

- CLT:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow^d Z$$

- When σ is unknown, we use the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

The Slutsky's theorem:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \rightarrow^d Z$$



$$H_0 : \mu = \mu_0$$

- The test statistics and its sampling distribution is

$$Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \rightarrow^d Z.$$



The SOP ensures that
 $P(\text{Type I error}) \leq \alpha$

THE TWO-SIDED TEST

1. $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$
2. Set up α
3. $Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$
4. Calculate the realized statistic Z^* .
5. Find (a) or (b)
 - (a) The rejection region = $\{z : z < -z_{\alpha/2}, z > z_{\alpha/2}\}$
 - (b) The p -value = $2P(Z > |Z^*|)$.
6. Conclude.
 - (a) If Z^* falls into the rejection region, we reject H_0 .
 - (b) If $p\text{-value} < \alpha$, we reject H_0 .



The SOP ensures that
 $P(\text{Type I error}) \leq \alpha$

THE LEFT-SIDED TEST

1. One of the following:

- $H_0 : \mu = \mu_0$ vs $H_a : \mu < \mu_0$
- $H_0 : \mu \geq \mu_0$ vs $H_a : \mu < \mu_0$
- $H_0 : \mu > \mu_0$ vs $H_a : \mu \leq \mu_0$

2. Set up α

$$3. Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$$

4. Calculate the realized statistic Z^* .

5. Find (a) or (b)

(a) The rejection region = $\{z : z < -z_\alpha\}$

(b) The p -value = $P(Z < Z^*)$.

6. Conclude.



The SOP ensures that
 $P(\text{Type I error}) \leq \alpha$

THE RIGHT-SIDED TEST

1. One of the following:

- $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$
- $H_0 : \mu \leq \mu_0$ vs $H_a : \mu > \mu_0$
- $H_0 : \mu < \mu_0$ vs $H_a : \mu \geq \mu_0$

2. Set up α

$$3. Z_{STAT} = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \xrightarrow{d} Z$$

4. Calculate the realized statistic Z^* .

5. Find (a) or (b)

(a) The rejection region = $\{z : z > z_\alpha\}$

(b) The p -value = $P(Z > Z^*)$.

6. Conclude.





EXAMPLE

- The daily yield for a chemical plant has averaged 880 tons for several years. The quality control manager wants to know if this average has changed. She randomly selects 50 days and records an average yield of 871 tons with a standard deviation of 21 tons.

$$H_0 : \mu = 880$$

$$H_a : \mu \neq 880$$

Test statistic :

$$z \approx \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{871 - 880}{21 / \sqrt{50}} = -3.03$$

EXAMPLE



To make our decision clear, we choose a significance level, say $\alpha = .01$

5. The rejection region is

$$\{z : z < -2.575 \text{ or } z > 2.575\}$$

6. Because $z^* = -3.03$ falls into the rejection region, we reject H_0

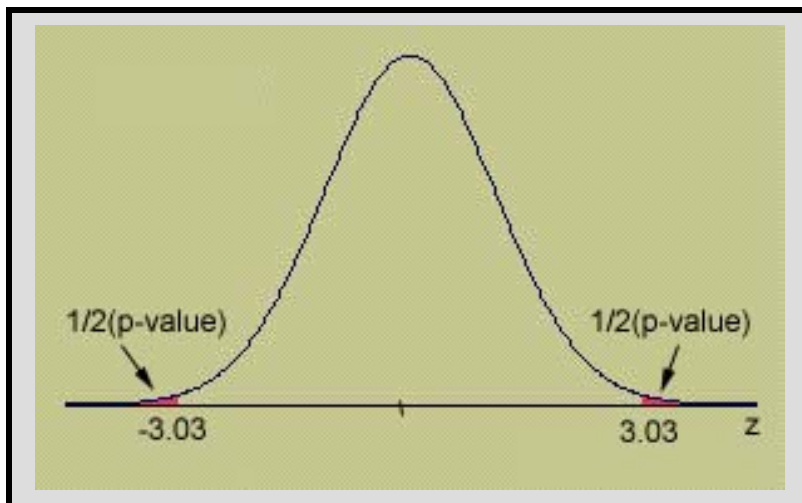




EXAMPLE

What is the probability that this test statistic or something even more extreme (far from what is expected if H_0 is true) could have happened *just by chance*?

$$\begin{aligned} p\text{-value} &: P(z > 3.03) + P(z < -3.03) \\ &= 2P(z < -3.03) = 2(.0012) = .0024 \end{aligned}$$



This is an unlikely occurrence, which happens about 2 times in 1000, assuming $\mu = 880$!

EXAMPLE



- To make our decision clear, we choose a significance level, say $\alpha = .01$.

If the p -value is less than α , H_0 is rejected as false. You report that the results are statistically significant at level α .

If the p -value is greater than α , H_0 is not rejected. You report that the results are not significant at level α .

Since our p -value = .0024 is less than α , we reject H_0 and conclude that the average yield has changed.





USING A REJECTION REGION

If $\alpha = .01$, what would be the **critical value** that marks the “dividing line” between “not rejecting” and “rejecting” H_0 ?

If $p\text{-value} < \alpha$, H_0 is rejected.

If $p\text{-value} > \alpha$, H_0 is not rejected.

The dividing line occurs when $p\text{-value} = \alpha$. This is called the **critical value** of the test statistic.

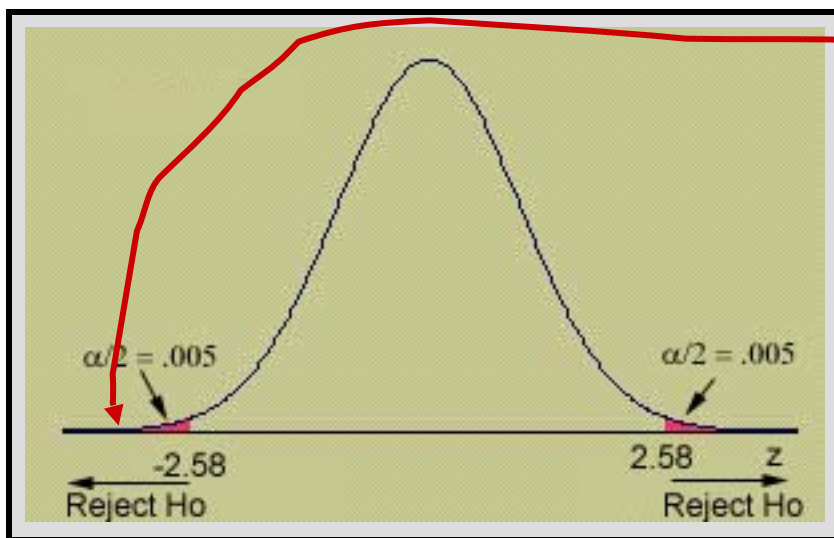
Test statistic $>$ critical value implies $p\text{-value} < \alpha$, H_0 is rejected.

Test statistic $<$ critical value implies $p\text{-value} > \alpha$, H_0 is not rejected.



EXAMPLE

What is the critical value of z that cuts off exactly $\alpha/2 = .01/2 = .005$ in the tail of the z distribution?



For our example, $z = -3.03$ falls in the rejection region and H_0 is rejected at the 1% significance level.

Rejection Region: Reject H_0 if $z > 2.58$ or $z < -2.58$. If the test statistic falls in the rejection region, its p -value will be less than $\alpha = .01$.



EXAMPLE

- A homeowner randomly samples 64 homes similar to her own and finds that the average selling price is \$252,000 with a standard deviation of \$15,000. Is this sufficient evidence to conclude that the average selling price is greater than \$250,000? Use $\alpha = .01$.



- Assume $X_i \sim \mathcal{G}(\mu, \sigma^2)$ for $i = 1, \dots, 64$.
- $\mathcal{G}$ is unknown.
- $\bar{x} = 252,000$, $s = 15,000$.



1. $H_0 : \mu = 250,000$ vs $H_1 : \mu > 250,000$

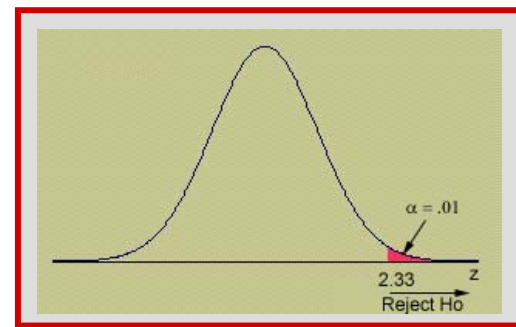
2. Set $\alpha = 0.01$

3. The test statistic is $Z_{STAT} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \rightarrow Z$.

$$4. Z^* = \frac{252000 - 250000}{15000/\sqrt{64}} = 1.07$$

5. Rejection region = $\{z : z > 2.33\}$

6. Because Z^* does not fall into RR. Do not reject H_0 .



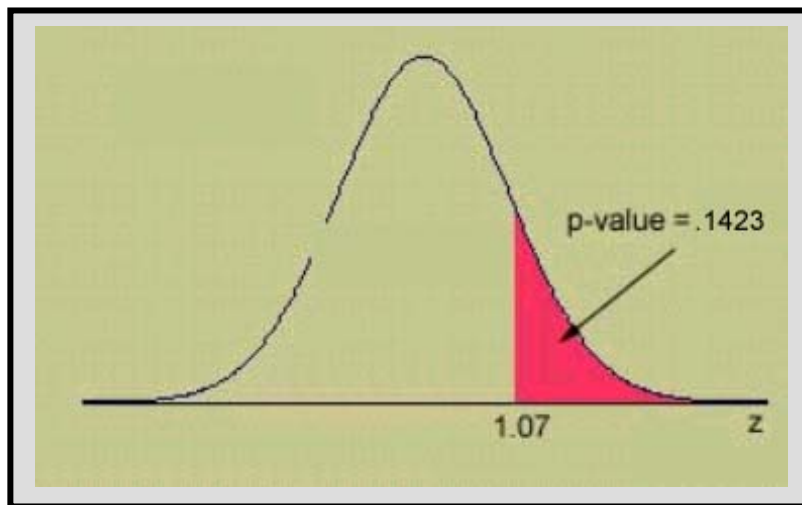
正面表述：Average price is likely about 250,000.

負面表述：There is not enough evidence to indicate that μ is greater than \$250,000.

P-VALUE APPROACH



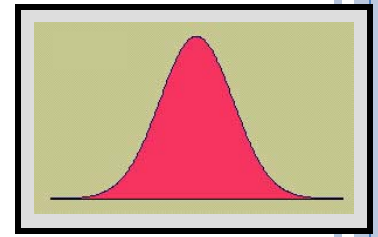
$$p\text{-value: } P(z > 1.07) = 1 - .8577 = .1423$$



Since the p -value is greater than $\alpha = .01$, H_0 is not rejected.

There is insufficient evidence to indicate that μ is greater than \$250,000.

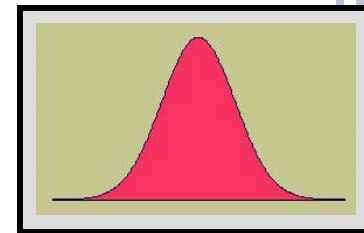
WE PREFER TO REPORT P-VALUE!



- The p -value approach is often preferred because
 - You can evaluate the test results at any significance level you choose.
- What should you do if you are the experimenter and no one gives you a significance level to use?



STATISTICAL SIGNIFICANCE



- If the p -value is less than **.01**, reject H_0 . The results are **highly significant**.
- If the p -value is between **.01** and **.05**, reject H_0 . The results are **statistically significant**.
- If the p -value is between **.05** and **.10**, do not reject H_0 . But, the results are **tending towards significance**.
- If the p -value is greater than **.10**, do not reject H_0 . The results are **not statistically significant**.



TWO TYPES OF ERRORS



We want to keep the probabilities of error as small as possible.

- **The value of α is the significance level, and is controlled by the experimenter.**
- **The value of β is difficult, if not impossible to calculate.**

Rather than “accepting H_0 ” as true without being able to provide a measure of goodness, we choose to “not reject” H_0 .

We write: There is insufficient evidence to reject H_0 .

The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue, arranged in a vertical line.

9.2 POWER OF A TEST (SKIPPED, SEE YOUTUBE VEDIOS)

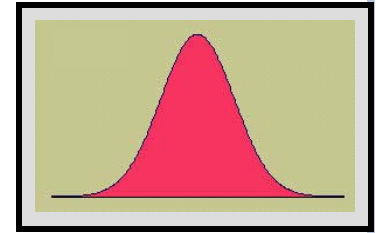


9.3 TWO POPULATION MEANS (SKIPPED 20251122)

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9.4 POPULATION PROPORTION

TESTING A BINOMIAL PROPORTION p



A random sample of size n from a binomial population to test

$H_0 : p = p_0$ versus

H_a : one of three alternatives

$$\text{Test statistic : } z \approx \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$$

with rejection regions and/or p - values based on the standard normal z distribution.





EXAMPLE

- Regardless of age, about 20% of American adults participate in fitness activities at least twice a week. A random sample of 100 adults over 40 years old found 15 who exercised at least twice a week. Is this evidence of a decline in participation after age 40? Use $\alpha = .05$.

$$H_0 : p = .2$$

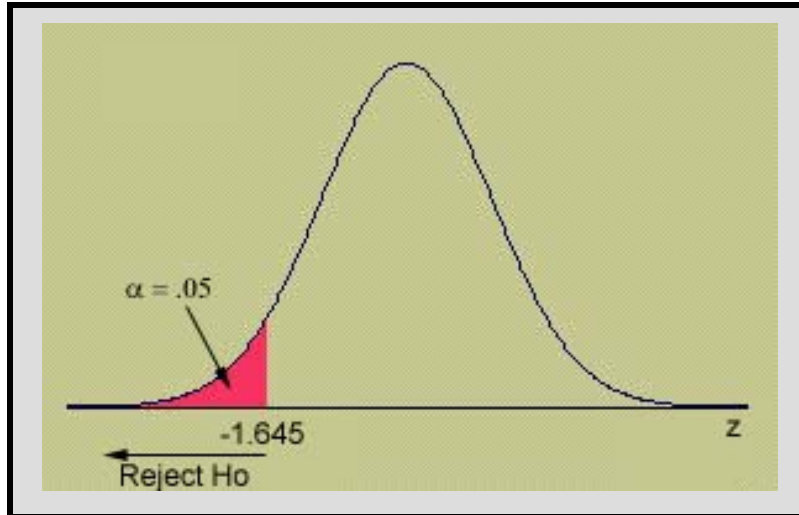
$$H_a : p < .2$$

Test statistic :

$$z \approx \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{.15 - .2}{\sqrt{\frac{.2(.8)}{100}}} = -1.25$$

CRITICAL VALUE APPROACH

What is the critical value of z that cuts off exactly $\alpha = .05$ in the left-tail of the z distribution?



For our example, $z = -1.25$ does not fall in the rejection region and H_0 is not rejected. There is not enough evidence to indicate that p is less than .2 for people over 40.

Rejection Region: Reject H_0 if $z < -1.645$. If the test statistic falls in the rejection region, its p -value will be less than $\alpha = .05$.

The left side of the slide features a series of vertical stripes in various shades of blue, ranging from light to dark. Overlaid on these stripes are several circles of different sizes, also in shades of blue, creating a modern, abstract design.

9.5 TWO POPULATION PROPORTIONS (SKIPPED 20251122)